MACH1 / FleetLine E1 — Powertrain & Technical Specs

Slide content pack · Items 3 & 4 · jury-facing

SLIDE 1 — POWERTRAIN & ENERGY STRATEGY

① The choice, and why it is not close

BEV. LFP. Single-speed reduction drive.

> *"For taxis, high annual mileage makes EVs the most cost-effective option, followed by CNG where infrastructure exists." — CEEW, cost-of-ownership study*

One qualifier earns the whole answer: our duty cycle is **200 km/day returning to a depot**. That is the single most favourable operating profile in the Indian market.

② Why 200 km/day decides it

- Fleet TCO parity with diesel emerges at **120–200 km/day**; payback in **18–24 months**
- Return-to-base parity can start at **100–150 km/day**
- **Our duty sits at the top of that band** — the margin is widest exactly where we operate
- *The Hindu BusinessLine, Feb 2026 — EKA Mobility, Euler Motors*

③ The number that actually matters: cost per km

Operator cost, 5 years, 312,000 km

	EV	CNG
Cost/km	**₹6.42**	₹7.63
Less residual	**₹4.82**	₹6.67
Advantage	**16% / 28%**	—

Robust to the fuel-economy assumption:

CNG mileage	20 km/kg	25 km/kg	30 km/kg
CNG cost/km	₹8.34	₹7.63	₹7.42
EV advantage	**23%**	**16%**	**13%**

> *"Higher kilometers traveled equates to better relative TCO of the EV option." — Cadmus / Uber, Electrifying Ridehail in India, Jan 2026*

④ Why charging is the *enabler*, not the objection

- **36 kWh overnight (10→90%) = 4.9 h at 7.4 kW · 3.3 h at 11 kW** → done before the shift
- **Zero revenue-hours lost to charging.** No queue, no offline, no missed target.
- Depot off-peak **₹5.50–9/kWh** vs public fast charge **₹18–24/kWh**
- Smart off-peak scheduling cuts charging cost **20–35%**; **1:3 charger-to-vehicle** is achievable

⑤ Where we are not claiming

- **Zero tailpipe — not zero emission.** India's grid is ~75% coal.
- CNG has better refuelling readiness. It is our **named fallback** if depot charging fails.
- EV's cost advantage is **geography-dependent** — it leads where charging exists.

SLIDE 2 — TECHNICAL SPECIFICATIONS

① Range and energy

- **45 kWh LFP** · 10–90% usable (36 kWh) · **263 km claim** — conservative
- **Verified consumption 12.84 kWh/100 km → real range ~315 km**
- **25.7 kWh/day** at 200 km/day · **0.63 cycles/day**
- Daily charge: **~36 kWh · 3.3 h at 11 kW overnight**

② Battery & energy system

- **PMSM, 80 bhp continuous / 100 bhp peak**, water-cooled
- **Single-speed reduction drive** · peak efficiency **93–97%** · ~2% loss at 45 °C
- **Active liquid-cooled pack**, held ≤45 °C
- **1–2C peak charge** (45 kW at 1C) · **overnight AC at 11 kW**
- Chemistry locked to **LFP** — no dependence on unproven sodium chemistry

③ Safety

- Airbags · ABS · **ESC** · Level 2 ADAS
- **AIS-156** thermal propagation · **AIS-038 Rev.2** EV powertrain safety
- **Microprocessor BMS** with cell-level voltage and temperature monitoring
- 72-hour edge buffer — critical safety events handled on-vehicle, not via cloud

④ Durability for commercial duty

- 62,400 km/yr · **312,000 km over the loan**
- **Drive unit 200,000+ km** rated — *3.2 years at our duty; replacement planned, not assumed away*
- **Battery first life ~15 years** (198 cycles/yr, not 365)
- **Service interval in months, not kilometres** — high-km vehicles otherwise over-serve worn parts
- Corrosion and vibration validated for monsoon and coastal duty

⑤ Comfort, interior and luggage

- **5-seat, 2-row, rear-biased** — paying passengers sit where it is roomiest
- **Flat floor, no tunnel** — EV architecture advantage
- **Rear AC vents** · ventilated seats · durable leatherette
- **400 L boot** — separate, unobstructed, matches airport-transfer demand

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SLIDE 3 — WHY WE CHOSE IT

① The alternatives, and why each loses

Option	Why not
CNG	Winner on refuelling readiness. **Loses on cost at our duty** — the capital premium takes ~5 years to repay. **Our fallback.**
Bio-CNG	Best carbon story of any fuel. **Not deliverable:** 207 plants live vs a 5,000 target; only **17 pipeline-connected**; plants running at 20–60% capacity.
HEV	**Taxed out at 43% GST + cess.** Also inherits the full ICE service burden while adding a battery.
PHEV	Same, plus two powertrains to service and two failure modes to train.
Mix	Our customer runs **one fleet, one service network, one training pipeline.** Maruti's edge *is* that single ecosystem.

② Sustainability — the honest version

- **Zero tailpipe emissions** in Delhi NCR, where air quality is the binding constraint and mandates are already gazetted
- **~50–60% below petrol** well-to-wheel today — improves as the grid cleans
- **A depot fleet can decarbonise its own electricity.** A gas fleet buys whatever the pipeline has.
- Bio-CNG would be lower-carbon still — which is why it stays on the watch list for post-2035

③ What we know we don't know

- **LFP calendar ageing at 40 °C is genuinely contested** — two peer-reviewed studies differ by **3.6×** on time-to-80%. We carry the full range rather than pick one.
- **Charging geography, not chemistry, drives regional adoption.** EV taxis lead where hubs exist.
- **Our duty assumption is the load-bearing input.** At 300 km/day a 45 kWh pack does not close the day — and we have sized for 200.

④ Why that honesty wins

Every number here is either measured on our duty cycle, taken from a named source, or labelled as an assumption. The pack sizing, the cost model and the charging plan all trace to one stated duty cycle — **200 km/day, return to base** — and we show what breaks if that number is wrong.

TRIM TRIAGE (if you need to cut)

Cut first — least load-bearing:

- Slide 3 ② sustainability bullets 3–4 (keep the tailpipe claim, drop the CNG methane detail if short)
- Slide 1 ⑤ "where we are not claiming" — retain only if there is room for a caveat slide
- Slide 2 ④ corrosion/vibration line

Never cut:

- Slide 1 ③ the cost-per-km table — it is the argument
- Slide 1 ④ the overnight-charging feasibility — it is the condition
- Slide 3 ① the alternatives table — "choose among BEV/HEV/PHEV/CNG/Bio-CNG" is an explicit ask
- Slide 3 ③ the contested-parameter line — it is the credibility anchor

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SPEAKER NOTES — five questions you will be asked

Q1. "Why not CNG? It's cheaper to buy and refuelling is everywhere."

> *Refuelling readiness is CNG's genuine win — I won't argue otherwise. But our duty is 200 km/day, which is exactly where the economics flip: EV wins 16–28% on cost per kilometre over five years, and it wins across every plausible CNG fuel-economy assumption. CNG becomes the fallback, not the answer, once depot charging is in place.*

Q2. "What happens when a charger is down?"

> *That's the wrong question for our operating model, and it's the right question for the public-charging fleet that failed in Noida. We charge at our own depot overnight. Three and a third hours at 11 kW, done before the shift starts. Zero revenue-hours exposed. We also carry CNG as a named fallback.*

Q3. "Is 45 kWh enough?"

> *Comfortably, and we are being conservative. Our claim is 263 km. On the verified duty cycle the same pack gives about 315 km. It covers our 200 km/day with margin. I should be clear that this holds at 200 km/day — at 300 km/day we would need roughly 51 kWh, and we have sized for the duty we actually run.*

Q4. "Why not Na-ion, or hybrids?"

> *Hybrids are taxed into uncompetitiveness at 43% GST plus cess, and they keep the entire ICE service burden while adding a battery. Na-ion we assessed seriously: it's blocked by an Indian certification standard that names lithium chemistry, not by cost. LFP is commercial, proven, and available now. We'd rather ship the chemistry that exists.*

Q5. "Your battery warranty claim?"

> *We separate first life from warranty, deliberately. First life on our duty cycle is roughly 15 years. The warranty is a smaller and less certain number, because two peer-reviewed LFP studies disagree by 3.6× on time-to-80% at 40 °C. We carry both, and we run a six-cell validation programme to close it. I'd rather show you the spread than pick the flattering end.*

CORRECTIONS TO THE EXISTING DECK

#	Issue	Fix
1	Table headed **"Total Cost of Ownership"** actually contains driver net take-home — higher is better	Relabel; add a separate operator cost/km table
2	EV take-home printed **₹16,380**; inputs give **₹16,360**	Correct to 16,360
3	Consumption stated as 15.38 kWh/100 km in places, 12.84 in others	Use 12.84 as primary, state basis
4	Duty cycle **200 km/day** (deck) vs **250–300 km/day** (survey)	Pick one; we recommend 200 + return-to-base
5	Cycles/year stated as **365**; actual at our duty is **~198**	Correct — pack life is ~15 yr, not 8.2
6	Boot **400 L** in spec table, **420 L** in graphic	One number only
7	**ESC absent** from safety spec	Add ESC
8	**Pack thermal management not specified**	Add active liquid cooling, pack ≤45 °C
9	HVAC line items sum to **14.80 kWh/day** vs 12.84 verified duty	Reconcile and state AC load explicitly
10	No statement of the operating premise	Add: return-to-base, overnight depot AC charging